

IDENTIFICATION

CODE : GI-3-S2-EC-GFL
ECTS : 2

HOURS

Cours : 0h
TD : 4h
TP : 24h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 28h
Travail personnel : 8h
Total : 40h

ASSESSMENT METHOD

Report and simulation model

TEACHING AIDS

Moodle

TEACHING LANGUAGE

French
English

CONTACT

M. VERCRAENE Samuel :
samuel.vercraene@insa-lyon.fr
M. MONTEIRO Thibaud :
thibaud.monteiro@insa-lyon.fr

AIMS

This course belongs to teaching unit Gestion de la chaîne logistique (GI-3-S2-UE-GECL) and contributes to the following skills :

- A1 Analyze a system (real or virtual) or a problem (Level 3)
- A2 Operate a model of a real or virtual system (Level 3)
- A3 Implement an experimental approach (Level 3)
- A6 Report on an analysis or on a scientific approach (Level 1)
- C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)
- C2 Modeling and designing an information, decision and production system, of goods and services (Level 3)
- C3 Evaluating, prototyping and simulating a system (Level 3)
- C4 Sizing the hardware and / or software of a system (Level 2)
- C9 Localizing and assigning the production, storage and transportation processes to different members of the supply chain (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Physical behavior of a chain of Actuators and Sensors
- To be able to :
 - (A1, A2, A3, A6, C1, C2, C3, C4, C9)
 - Apply a "simulation project" approach
 - Trace flows
 - Analyze average flows
 - Find the bottleneck
 - Make stocks functional
 - Sizing a production line
 - Propose modifications to optimize flows
 - Implement an experimental approach to validate the proposed modifications

CONTENT

Apply a "simulation project" approach

1. Define Performance Indicators
2. Find the data required for the simulation
3. Model the problem by simplifying it
4. Validate the model

Finding a Bottleneck

Propose changes to optimize flows

Implementing tools and fundamentals of industrial engineering in a simulation project: Stock management, Experimental design, Stochastic processes, Queueing systems...

BIBLIOGRAPHY

PRE-REQUISITES

Production management,
Experimental design,
Stochastic processes,
Queueing systems